Causal Steering of Protein Language Models: When Correlational Validation Fails, and How to Catch It

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Abstract

Activation steering has recently been shown to reproduce a known protein property (thermostability) in ESM2-650M via difference-of-means vectors. We ask whether a scoring proxy's correlation with real experimental labels predicts whether steering toward it actually works, and find the answer is no. Under a pre-registered six-criterion protocol testing five novel target properties, our best-validated proxy (binding affinity, $r = 0.80$ held-out) steers nothing, while a substantially weaker one ($r = 0.22$, catalytic activity) produces a robust, three-seed-replicated effect. This mirrors an independent, component-level finding in the same model family: attention heads ranked by correlation with real 3D contacts show near-zero relationship to their causal ablation effect. We show a single-seed steering result can mislead in criterion-specific ways – one target's effect direction replicates across three seeds while its residue-exclusion check does not (2 of 3) – and introduce a lightweight diagnostic, steering-vector cosine similarity across targets, that explains an otherwise-unexplained ambiguous result as a geometric echo of a different property's validated direction. We release the full protocol, all target harnesses, and the seed-replication data.

1 Introduction

Difference-of-means activation steering – adding a direction built from the mean activation difference between high- and low-property example groups to a residual stream – reproduces known thermostability biology in ESM2-650M [Huang et al., 2025]. We ask the natural follow-up: does a scoring proxy's correlation with real labels predict whether steering toward a *new* property works? Across five novel targets under one pre-registered protocol, the best-validated proxy in our study ($r = 0.80$, binding affinity) produces a flat null effect, while a substantially weaker proxy ($r = 0.22$, catalytic activity) produces a clean, three-seed-replicated effect. This mirrors an independent, component-level result: Vig et al. [Vig et al., 2021] found an attention head whose correlation with real 3D contacts is $12.9\times$ enrichment, explicitly flagged as untested causally; ablating it gives an effect indistinguishable from a low-correlation control, and a full 480-head sweep ranks the correlational top pick 313th of 480 by actual causal effect. Two levels, two techniques, one lesson: correlational signal is evidence about representations, not about intervention. Our contribution is the audit discipline needed to trust a causal claim here, demonstrated via two near-misses caught in our own pipeline before they could be reported as clean results.

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Table 1: Five novel targets. Proxy r: Pearson correlation vs. real labels, held-out, before any steering run.

Target	Proxy r	Steering result	Verdict
Binding affinity	0.80	flat null, all α	KILL
Catalytic activity	0.22	PASS, 3/3 seeds	**PASS**
Intrinsic disorder	0.45	3/3 direction; 2/3 robustness	**PASS***
Immunogenicity	$\leq$0.10	not run	KILL (pre-run)
Expression yield	0.31	fails residue-exclusion	AMBIGUOUS

*PASS on criteria 1,2,4,6, and 2/3 seeds on criterion 3.

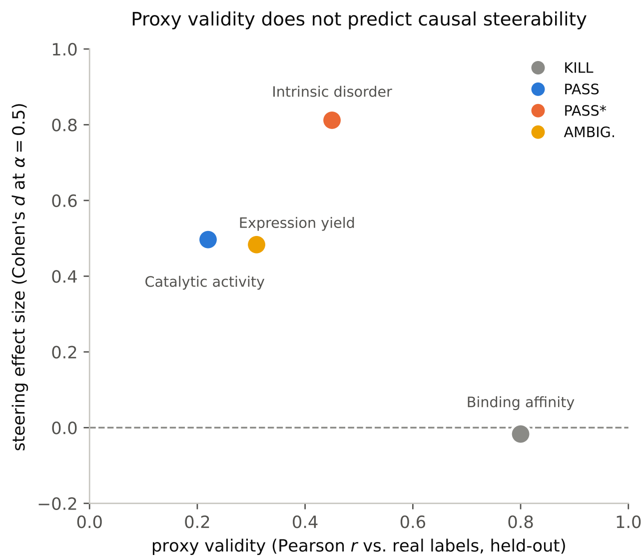

Figure 1: Proxy validity vs. steering effect size (Cohen's d, $\alpha = 0.5$). The best-validated proxy (binding affinity, $r = 0.80$) gives the smallest effect; a proxy less than half as strong (catalytic activity, $r = 0.22$) gives one of the two largest. No positive trend.

2 Method

We lock six criteria before testing any target, computed automatically: (1) the real direction beats a matched-norm random-direction control via paired bootstrap (10,000 resamples); (2) a monotonic dose-response across ≥ 3 steering strengths, restricted to a pre-established safe range (§4); (3) the effect survives residue-exclusion; (4) the proxy is validated against real labels *before* the run; (5) it beats the best existing technique where one exists; (6) ≥ 150 held-out evaluation sequences. A target failing any criterion outright is a **KILL**; one passing 1–2 but failing 3 or 4 is **AMBIGUOUS**.

3 The Central Finding: Proxy Validity Does Not Predict Causal Effect

Binding affinity (ProteinGym DMS, KRAS/DARPin): a mutational-sensitivity-weighted proxy validated more rigorously than any other target here ($r = 0.80$ held-out; weight-shuffle null $r = -0.001$; held-out-position generalization $r = 0.54$; single-to-double-mutant extrapolation $r = 0.70$) – yet the steering effect is a flat null at every α, zero degenerate sequences at any strength. A plausible mechanism: this dataset's vector-building groups are single-backbone point mutants (1–2 residues from a shared reference), giving difference-of-means little compositional signal to act on – measured directly, mean compositional distance between low/high groups is 0.003 here vs. 0.02–0.05 for the three cross-protein targets below.

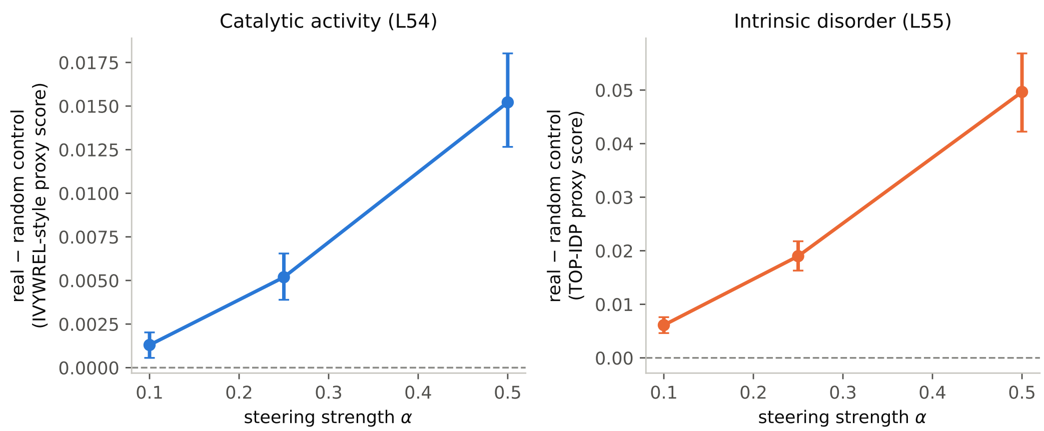

Figure 2: Dose-response, the two targets passing criterion 2. Both monotonic, 95% bootstrap CIs bounded away from zero throughout.

Catalytic activity (DLKcat, cross-protein kcat): a glycine-minus-arginine proxy, the weakest of the four runnable targets ($r = 0.22$), matching the published activity-stability tradeoff in enzymology. Produces a clean, monotonic, residue-robust effect (Figure 2, left) replicated identically across three independent seeds – the one target with fully unanimous seed-robustness.

Intrinsic disorder (DisProt, TOP-IDP scale, $r = 0.45$, the strongest proxy here, AUC 0.71 at the per-residue level): the effect is significant and monotonic on all 3 seeds (Figure 2, right), but its residue-exclusion check passes on 2/3 seeds (36–42% magnitude retained) and fails on the third (10% retained, CI crosses zero) – existence and direction are seed-stable; the artifact check is not, uniformly.

Immunogenicity (IEDB, 4 tiers): proxies predictive on a surrogate binding-affinity tier ($r = 0.37$–0.43) collapse near-chance on the real T-cell-response endpoint ($r \leq 0.10$); a full-length-antigen proxy reaching $r = 0.38$ is a source-organism confound (organism identity explains 58% of the apparent signal; flips to $r = -0.32$ once held out of cross-validation). Killed before any steering run.

Expression yield (eSol, net-charge-deviation proxy, $r = 0.31$–0.34): significant, dose-responsive, but collapses under residue-exclusion. Rather than discarding this, we compute steering-vector cosine similarity against every other target: $+0.30$ overall with intrinsic disorder's vector, rising to $+0.40$–0.50 at the three deepest layers – consistent with disordered regions independently reducing solubility/expression in the literature. This is best read as a partial echo of disorder's already-validated effect through a more collapse-prone proxy, not a sixth steerable property.

4 Self-Correction Case Studies

A spurious pass from an unsafe comparison. Testing whether restricting steering to the 5 layers previously found causally necessary preserves the full 33-layer effect, a first pass selected its best-performing α from the whole tested range, including $\alpha \geq 1.0$, and reported a clean pass at $\alpha = 2.0$. This was comparison artifact: at that α the 33-layer arm had already fully collapsed into degenerate output (zero non-degenerate sequences), while the 5-layer arm had not – it won only by breaking down later. Restricting the comparison to a range where neither arm has collapsed and rerunning end to end: the 5-layer subset is real but retains only 43–59% of the full effect.

A criterion passing on only 2 of 3 seeds. The disorder target's residue-exclusion check (§3) is seed-sensitive even though the headline effect is not – a single seed could have reported either a false unanimous pass or a false kill on this one criterion, undetectable without a multi-seed rerun.

5 Related Work

Huang et al. [2025] validate this steering method on thermostability; we extend it to five new properties under a pre-registered protocol and test, to our knowledge for the first time, whether proxy correlation strength predicts steering success across unrelated properties. Vig et al. [2021] identify the head we redo as a genuine causal test. Adams et al. [2025] train SAEs on ESM-2 and show via linear probing – correlational – that thermostability and localization determinants are identifiable in SAE feature space; their one steering check is a coherence test, not validation against a real label, and their discussion states "investigating how to steer the model in useful ways remains an open direction for future work." We answer a version of that for a non-SAE mechanism, specifically for whether correlational validity predicts causal usefulness.

6 Discussion

Every steering result here is causal about the model's activation space – a real intervention against a random-direction control and a residue-exclusion check – not validated against a wet-lab assay: a PASS means "causally steers a validated proxy," not "causally engineers the biological property." The vector-geometry check that explained the expression-yield result is cheap (one extra embedding pass) and worth running by default whenever multiple targets are tested together. All experiments use one model (ESM2-650M) at one scale; all proxies are compositional heuristics; seed-robustness was checked for two of five targets; no structural/functional plausibility check was run on generated sequences.

7 Conclusion

Correlational validity – of an attention head's biological alignment, or a proxy's correlation with real labels – does not predict causal effect when actually tested, at two levels of the same model class. Proxy validation strength is not a reliable filter for which targets are worth attempting, and a single steering run should not be trusted without a seed-robustness check where the result matters downstream.

Responsible-use statement

This work identifies methodological pitfalls in causally validating activation-steering claims on protein language models and does not introduce a new capability for generating harmful biological sequences; all target properties studied (thermostability, binding, catalysis, disorder, immunogenicity, expression) are common protein-engineering objectives with dual-use risk no greater than the underlying published steering method [Huang et al., 2025] we extend. We see no additional mitigation specific to this paper's contribution beyond standard biosecurity practice already governing protein-language-model research.

References

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